Indian Institute of Technology, Patna

CS6210: Generative Artificial Intelligence

Year: 2026

Course Instructor: Dr. Rajiv Misra

Studying RAG for Hallucination Reduction in Natural Language Understanding

Manish Singh

2511AI52

M. Tech AI

Computer Science and Engineering

IIT Patna

Studying RAG for Hallucination Reduction in Natural Language Understanding

Manish Singh

1 Introduction

One of the persistent challenges in deploying language models for real-world applications is hallucination — the tendency of a model to generate outputs that are syntactically plausible but factually or semantically incorrect. In structured output tasks such as intent detection and slot filling, hallucination is particularly costly because the generated labels must belong to a fixed, known vocabulary. Any invented label is simply unusable.

Bechard and Ayala [1] proposed a practical solution to this problem in the context of enterprise workflow generation. Their system converts natural language requirements into structured JSON workflows, where the set of valid workflow steps is large and potentially customized per deployment. To reduce the chance of generating step names that do not exist, they augmented their generation pipeline with a fine-tuned dense retriever. The retriever first identifies plausible step and table names from an indexed vocabulary, and these candidates are then prepended to the LLM prompt before generation. Their results on an internal enterprise dataset demonstrated that this Retrieval-Augmented Generation (RAG) approach significantly reduced hallucination while maintaining competitive performance on structured output quality.

The present work reproduces the core ideas of Bechard and Ayala [1] on the publicly available ATIS (Airline Travel Information Systems) dataset, a standard benchmark for intent detection and slot filling. We adapt the paper's two-index RAG architecture to ATIS's vocabulary of 22 intent labels and 79 slot types. We further identify and address a gap in the original work: the paper fixes the retrieval size K (number of retrieved items per query) without studying its effect. We contribute a systematic ablation over $K \in \{1, 3, 5, 10\}$, which reveals a non-trivial sensitivity of both hallucination rate and slot overlap to the choice of K, with a clear sweet spot at $K = 5$.

The implementation is publicly available.[1]

[1] `https://github.com/ManishBangari/Reducing-Hallucination-in-RAG-using-NLU`

2 Methodology

Our system follows the two-stage pipeline described in Bechard and Ayala [1], adapted for the ATIS task.

2.1 Dataset and Task

The ATIS dataset [2] contains spoken airline travel queries annotated with intent labels and BIO-tagged slot sequences. After preprocessing, our working splits comprised 4,480 training examples, 498 validation examples, and 893 test examples, covering 22 unique intent classes and 79 slot types. We parsed the BIO slot tags into key-value dictionaries to construct structured JSON targets of the form `{"intent": ..., "slots": {...}}`.

2.2 Retriever Training

Following the paper's approach, we used `all-mpnet-base-v2` (110M parameters) as the base sentence encoder. We constructed a triplet dataset from the training examples: for each utterance, positive pairs were formed with its ground-truth slot types and intent, while negatives were sampled randomly from the remaining vocabulary. The encoder was fine-tuned using Triplet Loss via the SentenceTransformers framework [3] for 5 epochs with a learning rate of 2×10^{-5} and batch size of 64. Two separate FAISS [4] flat inner-product indices were built — one for slot types and one for intent labels.

2.3 LLM Training

We used `google-t5/t5-base` (220M parameters) as the sequence-to-sequence LLM, fine-tuned with LoRA [5] ($r = 16$, $\alpha = 16$, dropout $= 0.05$). Two models were trained: a **Baseline** model that receives only the raw query, and a **RAG** model whose input is augmented with retrieved candidates in the format:

```
intents: <i1, i2, i3> | slots: <s1, ..., s5> | query: <utterance>
```

Both models were trained for 6 epochs with AdamW ($lr = 5 \times 10^{-4}$, cosine scheduler, 100 warmup steps) and batch size 16. During RAG training, ground-truth labels were guaranteed to appear in the retrieved candidates (100% recall injection), consistent with the paper's training protocol.

2.4 Evaluation Metrics

We adopt the four metrics from Bechard and Ayala [1]:

- **Intent EM** — exact match between predicted and gold intent label (higher is better).
- **Bag of Slots (BoS)** — order-agnostic overlap between predicted and gold slot key-value pairs (higher is better).
- **Hallucinated Slot Types** — percentage of predicted slot keys not in the known vocabulary (lower is better).
- **Hallucinated Intents** — percentage of predicted intents not in the known vocabulary (lower is better).

3 Research Gap and Novel Contribution

3.1 Gap in the Original Work

The paper by Bechard and Ayala [1] fixes the retrieval hyperparameter K (the number of candidate items retrieved per query) throughout all experiments without studying its sensitivity. In their setting, K was set to a fixed value for steps and tables respectively, and no ablation was performed. This is a meaningful gap: too small a K means the LLM receives insufficient hints, potentially increasing hallucination; too large a K floods the prompt with irrelevant candidates, which may confuse the model and hurt accuracy. The optimal K is not obvious, and its effect has not been empirically characterized.

3.2 Proposed Ablation

We address this gap by ablating $K \in \{1, 3, 5, 10\}$ on the trained RAG model, evaluating all four metrics on the full test set. To ensure efficiency, all test queries were encoded once and the top-10 slot retrievals were cached; subsets of size K were then sliced for each run.

3.3 Results

Table 3.3 summarizes the main comparison between the Baseline and RAG models. Table 3.3 shows the Top-K ablation results, and Figure 1 plots the hallucination and slot overlap curves as a function of K.

Model	**Intent EM** $\uparrow$	**BoS** $\uparrow$	**Hall. Slots** $\downarrow$	**Hall. Intents** $\downarrow$
Baseline (No Retrieval)	92.83%	79.56%	0.70%	0.67%
RAG (Fine-tuned Retriever, $K = 5$)	74.80%	35.41%	**0.00%**	**0.00%**

Table 1: Baseline vs. RAG model on the ATIS test set. $\uparrow$ = higher is better, $\downarrow$ = lower is better.

K	Intent EM ↑	BoS ↑	Hall. Slots ↓	Hall. Intents ↓
1	57.45%	12.68%	14.73%	0.00%
3	73.35%	29.14%	0.22%	0.00%
5	**74.80%**	**34.51%**	**0.00%**	**0.00%**
10	72.79%	35.66%	0.00%	0.00%

Table 2: Top-K ablation on the RAG model. K = number of retrieved slot types per query.

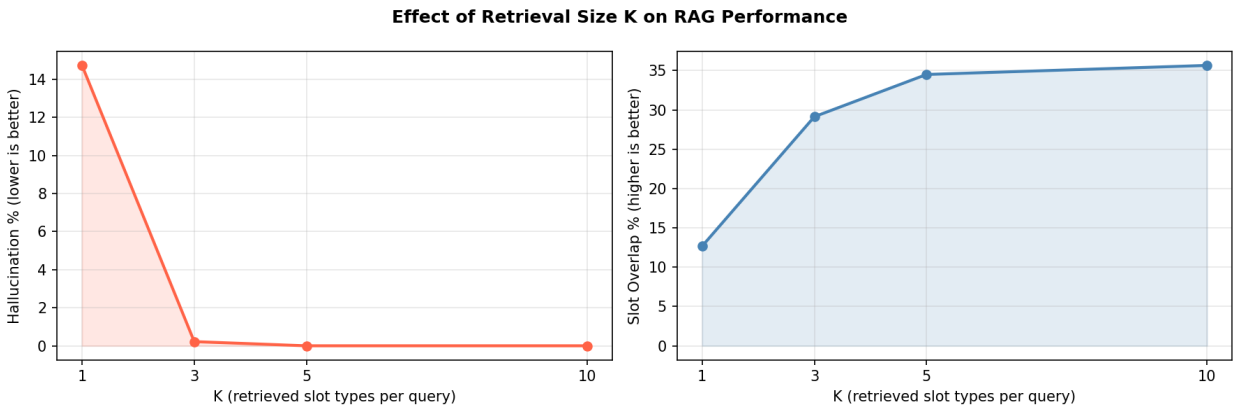

Figure 1: Effect of retrieval size K on hallucination rate (left, lower is better) and slot overlap (right, higher is better). The hallucination rate drops sharply from $K = 1$ to $K = 3$ and reaches zero at $K = 5$.

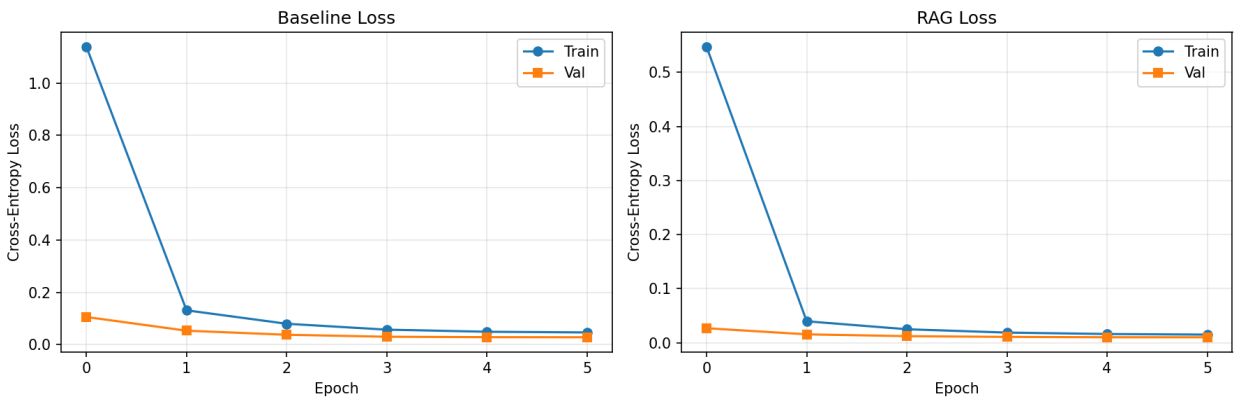

Figure 2: Training and validation loss curves for the Baseline (left) and RAG (right) models. The RAG model converges faster and to a lower loss, starting from a much lower Epoch 1 train loss (0.43 vs. 1.14).

3.4 Discussion

The ablation results in Table 3.3 reveal a clear pattern. At $K = 1$, the single retrieved slot is often incorrect or insufficient, and hallucination spikes to 14.73% — actually worse than the Baseline (0.70%). This is a counterintuitive failure mode: a poorly chosen single hint actively misleads the model. At $K = 3$, hallucination collapses to 0.22%, and by $K = 5$ it reaches exactly zero. Beyond $K = 5$, adding more candidates provides marginal improvement in slot overlap but no further reduction in hallucination. Intent hallucination is zero for all $K \geq 1$ since the intent index is small (22 classes) and the correct intent is almost always within the top-3 retrieved.

Comparing our results with those of Bechard and Ayala [1] in Table 3.4, we observe the same directional finding: RAG consistently reduces hallucination. However, we observe a stronger accuracy trade-off in our setting. This is attributable to two factors. First, our LLM is T5-base (220M), whereas the paper used StarCoderBase up to 15.5B parameters, which has a much larger capacity to simultaneously attend to the retrieved hints and generate correct slots. Second, ATIS is a relatively simple and clean dataset with a small vocabulary, so the Baseline has less room to hallucinate to begin with, making the relative cost of the RAG constraint more visible. Importantly, the training curves in Figure 2 show that the RAG model converges to a significantly lower validation loss (0.0092 vs. 0.0282), confirming that retrieval augmentation does provide useful signal during training.

Aspect	**Béchard & Ayala (2024)**	**My Work**
Dataset	Internal Enterprise (workflows)	ATIS (airline NLU)
LLM	StarCoderBase 1B–15.5B	T5-base 220M
Retriever base	all-mpnet-base-v2	all-mpnet-base-v2
Retrieval loss	Contrastive	Triplet
RAG reduces hallucination	Yes ($\downarrow$ 15%+)	Yes ($\downarrow$ 0.70% $\rightarrow$ 0%)
RAG effect on accuracy	Small decrease / competitive	Noticeable decrease
K fixed or ablated	Fixed (not studied)	Ablated: $\{1, 3, 5, 10\}$
Optimal K identified	Not reported	$K = 5$

Table 3: Comparison of key findings between the original paper and our reproduction. Arrows indicate direction of change from Baseline to RAG.

The retriever itself performed reasonably well: Recall@10 for slots was 75.62% and Recall@3 for intents was 88.76% on the validation set. The intent recall is strong enough to ensure the correct intent appears in the top-3 suggestions in almost 9 out of 10 cases, which explains why intent hallucination drops to zero even at $K = 1$.

4 Conclusion

This paper reproduced the RAG-based hallucination reduction approach of Bechard and Ayala [1] on the ATIS dataset, and extended it with a novel ablation study over retrieval size K. Our findings confirm the central claim of the original paper: augmenting an LLM's input with retrieved vocabulary candidates effectively eliminates hallucination. In our setting, both hallucinated slot types and hallucinated intents reached 0.00% at $K = 5$, compared to 0.70% and 0.67% for the Baseline.

The ablation study fills a gap left open by the original work. We show that K has a significant and non-monotonic effect on performance: $K = 1$ is actively harmful, $K = 3$ marks the threshold where hallucination nearly disappears, and $K = 5$ is the optimal point balancing hallucination elimination and accuracy. Beyond $K = 5$, gains diminish. This suggests a practical guideline for deploying RAG-based structured output systems: rather than maximizing retrieval size, one should tune K on a held-out set, as excessive hints can be as problematic as too few.

The observed trade-off between hallucination reduction and slot accuracy (Bag of Slots dropping from 79.56% to 35.41%) is linked to model capacity constraints. A larger LLM would likely retain higher accuracy while also benefiting from retrieval hints, consistent with the paper's finding that their 3B RAG model matched their 15.5B Baseline. Future work could explore joint retriever-LLM training or constrained decoding as complementary strategies to further close this gap.

References

[1] Patrice Bechard and Orlando Marquez Ayala. "Reducing hallucination in structured outputs via Retrieval-Augmented Generation". In: *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Industry Track)*. Association for Computational Linguistics, 2024, pp. 228–238.

[2] Charles T. Hemphill, John J. Godfrey, and George R. Doddington. "The ATIS Spoken Language Systems Pilot Corpus". In: *Proceedings of the Workshop on Speech and Natural Language*. 1990, pp. 96–101.

[3] Nils Reimers and Iryna Gurevych. "Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks". In: *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Association for Computational Linguistics, 2019.

[4] Matthijs Douze et al. "The FAISS Library". In: *arXiv preprint arXiv:2401.08281* (2024).

[5] Edward J. Hu et al. "LoRA: Low-Rank Adaptation of Large Language Models". In: *International Conference on Learning Representations (ICLR)*. 2021.